

# The recursion method of a linear operator inversion: III

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**Abstract.** We suggest constructing an inverse of the infinite-dimensional matrix  $Q$  by means of its recurrent algebraic decomposition into the easily invertible two-diagonal factors. The merits and feasibility of the method are illustrated on the five-diagonal matrices. Quartic and decadic–decadic anharmonic propagators are chosen as examples of the application.

## 1. Introduction

In quantum mechanics, manipulations with matrices (Hamiltonians, transition operators, etc) are usually facilitated by their diagonalisation. Limitations of this technique may be both computational and interpretational in character. As a consequence, the direct use of the next, tridiagonal form of matrices is also quite frequent and may affect both the approximation schemes (Bassichis and Strayer 1978 etc) and the construction of models (e.g. Haydock and Kelly 1975).

In the former context, one of the important characteristics of the infinite tridiagonal matrices is their easy invertibility in terms of continued fractions (Wall 1948). In paper I of the present series (Znojil 1976) we have, therefore, weakened the tridiagonality restriction and succeeded in inverting algebraically even the general Hessenberg matrix  $Q_{ij}$  (which is zero for all  $j \geq i + 2$ ) in terms of the auxiliary 'extended continued fractional' (ECF) sequence  $f_k$ . These ECF quantities were defined by the recurrences

$$f_k(N) = \left( a_k^{(1)} + \sum_{j=2}^{N-k+1} a_k^{(j)} \prod_{i=1}^{j-1} f_{k+i}(N) \right)^{-1} \quad (1.1)$$

$$k = 1, 2, \dots, N \quad f_{N+1}(N) = f_{N+2}(N) = \dots = 0$$

in the limit  $N \rightarrow \infty$  ( $f_k = \lim_{N \rightarrow \infty} f_k(N)$ ), i.e. totally analogous with the classical continued fractions.

In paper II (Znojil 1978) we used the generalised (matrix) form of recurrences (1.1) to invert any sparse matrix  $Q$  by means of its Hessenberg-type partitioning. In this paper, we intend to suggest an alternative generalisation of I which will be based on a representation of  $Q$  in the form of a product of the two-diagonal matrices.

For the sake of brevity, we consider in detail only the simplest non-trivial case, namely the matrix  $Q$  given in the five-diagonal form. Thus, the method is presented in § 2, while in § 3 the convergence and its acceleration are discussed thoroughly in the ECF context. Finally, § 4 is devoted to physical examples (anharmonic propagators), and the various possible generalisations of the formalism are reviewed in § 5.

## 2. The method

### 2.1. Five-diagonal matrices

Any finite five-diagonal matrix  $Q$

$$\begin{pmatrix} Q_{11} & Q_{12} & Q_{13} & 0 & \dots & & & & & 0 \\ \dots & & & & & & & & & \\ 0 & \dots & 0 & Q_{kk-2} & Q_{kk-1} & Q_{kk} & Q_{kk+1} & Q_{kk+2} & 0 & \dots & 0 \\ \dots & & & & & & & & & & \\ 0 & \dots & & & & & 0 & Q_{NN-2} & Q_{NN-1} & Q_{NN} \end{pmatrix} \quad (2.1)$$

may be inverted numerically by the algorithms described by Grund (1977). Alternatively, it may also be partitioned (into the blockwise tridiagonal form) and inverted in terms of the  $(2 \times 2)$ -dimensional matrix continued fractions (MCF) (Znojil 1977).

In this paper, we intend to construct  $Q^{-1}$  algebraically, by means of its complete factorisation into the easily invertible two-diagonal matrices

$$\begin{pmatrix} 1 & \alpha_1 & & & \\ & 1 & \alpha_2 & & \\ & & \ddots & \ddots & \\ & & & \ddots & \ddots \end{pmatrix} = \begin{pmatrix} 1 & -\alpha_1 & \alpha_1\alpha_2 & -\alpha_1\alpha_2\alpha_3 & \dots \\ & 1 & -\alpha_2 & \alpha_2\alpha_3 & \dots \\ & & \ddots & \ddots & \ddots \end{pmatrix}^{-1}. \quad (2.2)$$

In the 'non-degenerate' cases with  $Q_{k-2k} \neq 0$  and  $Q_{kk+2} \neq 0$ , we shall write

$$Q = \begin{pmatrix} G_1 & & & \\ & \ddots & & \\ & & G_N \end{pmatrix} \times H \times \begin{pmatrix} D_1 & & & \\ & \ddots & & \\ & & D_N \end{pmatrix}$$

$$H = \begin{pmatrix} a_1 & b_1 & 1 & \dots & & \\ \dots & & & & & \\ \dots & 1 & c_k & a_k & b_k & 1 & \dots \\ \dots & & & & & & \\ \dots & & & & 1 & c_N & a_N \end{pmatrix} \quad (2.3)$$

$$Q_{kk-2} = G_k D_{k-2} \quad Q_{kk+2} = G_k D_{k+2}$$

with the trivial recurrences defining

$$\begin{aligned} D_3 &= Q_{13}/G_1 & D_4 &= Q_{24}/G_2 \dots \\ G_3 &= Q_{31}/D_1 & G_4 &= Q_{42}/D_2 \dots \end{aligned} \quad (2.4)$$

from the first four free values  $G_1$ ,  $G_2$ ,  $D_1$  and  $D_2$ . This simplifies the further notation—we may put

$$H = \begin{pmatrix} 1 & \alpha_1 & & & \\ & \ddots & \ddots & & \\ & & 1 & \alpha_{N-1} & \\ & & & 1 & \\ & & & & 1 \end{pmatrix} \times \begin{pmatrix} 1 & \beta_1 & & & \\ & \ddots & \ddots & & \\ & & 1 & \beta_{N-1} & \\ & & & 1 & \\ & & & & 1 \end{pmatrix} \times \begin{pmatrix} 1/f_1 & & & & \\ & \ddots & & & \\ & & & 1/f_{N-1} & \\ & & & & 1/f_N \end{pmatrix}$$

$$\times \begin{pmatrix} 1 & & & & \\ \gamma_2 & 1 & & & \\ & \ddots & \ddots & & \\ & & \gamma_N & 1 & \end{pmatrix} \times \begin{pmatrix} 1 & & & & \\ \delta_2 & 1 & & & \\ & \ddots & \ddots & & \\ & & \delta_N & 1 & \end{pmatrix} \quad (2.5)$$

and obtain the required result in the form

$$Q^{-1} = \begin{pmatrix} D_1^{-1} & & \\ & \ddots & \\ & & D_N^{-1} \end{pmatrix} \times H^{-1} \times \begin{pmatrix} G_1^{-1} & & \\ & \ddots & \\ & & G_N^{-1} \end{pmatrix} \quad (2.6)$$

$$H^{-1} = \begin{pmatrix} 1 & & & \\ -\delta_2 & 1 & & \\ \delta_2\delta_3 & -\delta_3 & 1 & \\ \dots & & & \ddots \end{pmatrix} \times \begin{pmatrix} 1 & & & \\ -\gamma_2 & 1 & & \\ \gamma_2\gamma_3 & -\gamma_3 & 1 & \\ \dots & & & \ddots \end{pmatrix} \times \begin{pmatrix} f_1 & & & \\ & f_2 & & \\ & & \ddots & \\ & & & \ddots \end{pmatrix} \\ \times \begin{pmatrix} 1 & -\beta_1 & \beta_1\beta_2 & \dots \\ & 1 & -\beta_2 & \dots \\ & & \ddots & \ddots \end{pmatrix} \times \begin{pmatrix} 1 & -\alpha_1 & \alpha_1\alpha_2 & \dots \\ & 1 & -\alpha_2 & \dots \\ & & \ddots & \ddots \end{pmatrix}.$$

In this way, the problem of inversion becomes reduced to the problem of the algebraic factorisation (2.5).

## 2.2. Recurrent factorisation

In (2.5), the  $2(N-2)$  elements of the two outer (unit) diagonals of  $H$  are represented by the products

$$1 = \alpha_k \beta_{k+1} / f_{k+2} \quad 1 = \delta_{k+1} \gamma_{k+2} / f_{k+2} \quad k = 1, 2, \dots, N-2. \quad (2.7)$$

When we denote  $u_k = \alpha_k + \beta_k$  and  $v_{k+1} = \gamma_{k+1} + \delta_{k+1}$ , we may re-express also the  $2(N-1)$  elements  $b_k$  and  $c_{k+1}$  of  $H$  in the form

$$b_k = u_k / f_{k+1} + \alpha_k \beta_{k+1} v_{k+2} / f_{k+2} \\ c_{k+1} = v_{k+1} / f_{k+1} + u_{k+1} \delta_{k+1} \gamma_{k+2} / f_{k+2} \quad k = 1, 2, \dots, N-1. \quad (2.8)$$

Finally, the main diagonal of  $H$  is related to the right-hand side of (2.5) by the relations

$$a_k = \frac{1}{f_k} + \frac{u_k v_{k+1}}{f_{k+1}} + \frac{\alpha_k \beta_{k+1} \delta_{k+1} \gamma_{k+2}}{f_{k+2}} \quad k = 1, 2, \dots, N-1. \quad (2.9)$$

Since  $b_N = \alpha_N = \beta_N = 0$  and  $c_{N+1} = \gamma_{N+1} = \delta_{N+1} = 0$ , we may initialise the recurrences (2.7) by any  $\alpha_{N-1} (= u_{N-1} - \beta_{N-1})$  and  $\delta_N (= v_N - \gamma_N)$  and define all the four two-diagonal factors in (2.5) by the recurrences

$$\alpha_k = f_{k+2} / (u_{k+1} - \alpha_{k+1}) \quad \delta_{k+1} = f_{k+2} / (v_{k+2} - \delta_{k+2}) \quad k = 1, 2, \dots, N-2. \quad (2.10)$$

Similarly, the fundamental sequences  $u_k$ ,  $v_{k+1}$  and  $1/f_{k+2}$ ,  $k = 1, 2, \dots, N-1$ , ( $1/f_{N+1} = 0$ ) may be generated from the trivial initialisation

$$u_N = v_{N+1} = f_{N+1} = 0 \quad f_N = 1/a_N. \quad (2.11)$$

The corresponding recurrences

$$u_k = f_{k+1} (b_k - v_{k+2}) \quad v_{k+1} = f_{k+1} (c_{k+1} - u_{k+1}) \quad (2.12)$$

and

$$f_k = 1/(a_k - u_k v_{k+1} / f_{k+1} - f_{k+2}) \quad k = N-1, N-2, \dots, 1 \quad (2.13)$$

are coupled and follow from (2.8) and (2.9), respectively. This completes our description of the numerical recurrent algorithm. We have only to perform the multiplication in (2.6),

$$H^{-1} = \begin{pmatrix} f_1, & -(\alpha_1 + \beta_1)f_1, & (\alpha_1\alpha_2 + \beta_1\beta_2 + \beta_1\alpha_2)f_1, & \dots \\ -(\gamma_2 + \delta_2), & (\alpha_1 + \beta_1)(\alpha_2 + \beta_2)f_1 + f_2, & \dots & \\ \dots & & & \end{pmatrix} \quad (2.14)$$

and obtain the inverse matrix for any finite  $N < \infty$ .

Let us note that the algorithm has the following two merits.

(1) Inverses of the lower submatrices of  $H$

$$H_{[M]} = \begin{pmatrix} a_M & b_M & 1 & \dots \\ \dots & & & \\ \dots & & 1 & c_N & a_N \end{pmatrix} \quad (2.15)$$

are obtainable, by a mere change of indices

$$H_{[M]}^{-1} = \begin{pmatrix} 1 & & & \\ -\delta_{M+1} & 1 & & \\ \dots & & & \\ \dots & & & 1 \end{pmatrix} \times \dots \times \begin{pmatrix} 1 & -\alpha_M & \dots & \\ & 1 & -\alpha_{M+1} & \dots \\ & & \dots & \\ & & \dots & 1 \end{pmatrix}, \quad (2.16)$$

as byproducts of our recurrent procedure. This may prove to be useful in the applications of the type described in § 4 below.

(2) The upper submatrices of the inverse  $H^{-1}$  have a simple form (2.14) where  $\alpha, \beta, \gamma$  and  $\delta$  may systematically be replaced by  $u$  and  $v$  only,

$$H^{-1} = \begin{pmatrix} f_1, & -u_1f_1, & (u_1, u_2 - f_3)f_1, & \dots \\ -v_2f_1, & (a_1 - f_3)f_2f_1, & (a_1b_2 + c_2 - a_1u_3)f_2f_2f_3, & \dots \\ \dots & & & \end{pmatrix} \quad (2.17)$$

(see appendix for details).

### 2.3. The $N \rightarrow \infty$ limit

Provided that the sequences  $u_k, v_{k+1}$  and  $f_{k+2}$  are known in the infinite-dimensional limit  $N \rightarrow \infty$ , the values of  $\alpha_k$  or  $\delta_{k+1}$  are given by their respective analytic continued fractional expansions

$$\alpha_k = \frac{f_{k+2}}{u_{k+1} - \frac{f_{k+3}}{u_{k+2} - \dots}}, \quad \delta_{k+1} = \frac{f_{k+2}}{v_{k+2} - \frac{f_{k+3}}{v_{k+3} - \dots}}. \quad (2.18)$$

In the light of the definitions of  $u_k$  and  $v_{k+1}$  we may write also, alternatively,

$$\beta_k = u_k - \frac{f_{k+2}}{u_{k+1} - \frac{f_{k+3}}{u_{k+2} - \dots}}, \quad \gamma_{k+1} = v_{k+1} - \frac{f_{k+2}}{v_{k+2} - \frac{f_{k+3}}{v_{k+3} - \dots}} \quad (2.19)$$

and apply the standard convergence criteria (Wall 1948).

In a similar way, sequences  $u_k$  and  $v_{k+1}$  may be defined directly in terms of the only sequence  $f_k$

$$\begin{aligned} u_k &= f_{k+1}b_k - f_{k+1}f_{k+2}c_{k+2} + f_{k+1}f_{k+2}f_{k+3}b_{k+2} - \dots \\ v_{k+1} &= f_{k+1}c_{k+1} - f_{k+1}f_{k+2}b_{k+1} + f_{k+1}f_{k+2}f_{k+3}c_{k+3} - \dots \end{aligned} \quad (2.20)$$

(see (2.12)). The appropriate convergence criteria should be used again (e.g. Korn and Korn 1968).

In the third step, the recurrence (2.13) may be reformulated in a surprising analogy with I.

**Theorem.** The diagonal matrix elements  $1/f_k$  in (2.5) coincide with the ECF quantities of the form (1.1).

*Proof.* We may consider only  $k = 1$  and  $N = \infty$  without loss of generality. Then, to obtain the expansion of  $f_1$  ( $=H_{11}^{-1}$ , cf (2.14)), we have to employ (2.13) with  $k = 2$  as a redefinition of

$$f_3 = -f_2f_3f_4 + a_2f_2f_3 - u_2f_2v_3$$

and insert it into (2.13) with  $k = 1$ . After an easy rearrangement (cf (2.12) with  $k = 1$  and (2.20)), we get precisely formula (1.1) where  $k = 1$  and the compact prescription

$$\begin{aligned} a_1^{(1)} &= a_1 & a_1^{(2)} &= -b_1c_2 & a_1^{(3)} &= b_1b_2 + c_2c_3 - a_2 \\ a_1^{(4)} &= -b_1c_4 - c_2b_3 + 1 & a_1^{(2n-1)} &= b_1b_{2n-2} + c_2c_{2n-1} \\ a_1^{(2n)} &= -b_1c_{2n} - c_2b_{2n-1} & n &\geq 3 \end{aligned} \quad (2.21)$$

defines the coefficients.

In the final step, the matrix elements of  $H^{-1}$  are to be specified as the sums of products of the ECF quantities  $f_k$  (further algebra is done in the appendix).

### 3. Fixed point initialisations

#### 3.1. Ambiguities of the inversion

In the infinite-dimensional cases, we may construct the non-trivial solutions  $w_k = -w_{k-1}/\alpha_{k-1}$ ,  $k = 2, 3, \dots$  to the homogeneous systems of equations

$$\begin{pmatrix} 1 & \alpha_1 & & \\ & 1 & \alpha_2 & \\ & & \ddots & \ddots \end{pmatrix} \begin{pmatrix} w_1 \\ w_2 \\ \vdots \end{pmatrix} = 0 \quad (3.1)$$

which can even be normalised for  $|\alpha_k| \geq 1 + \varepsilon$ ,  $\varepsilon > 0$ ,  $k > k_0$ . Therefore, up to some exceptional situations where  $w_1 \neq 0$  (Znojil 1983) we must complement (2.2) by the condition  $w_1 (= w_2 = \dots) = 0$ .

Furthermore, the different formal inverses  $H^{-1}$  may also be generated by the various auxiliary sequences, i.e. by the *a priori* unrestricted free choice of the four initial values

$$u_M \quad v_{M+1} \quad f_{M+1} \quad f_{M+2} \quad (3.2)$$

at some fixed  $M \geq 1$ . Hence, the choice of the values (3.2) represents, in fact, an independent 'boundary-condition-type' requirement. It has to suppress completely the algebraic ambiguity of the formal inverse (2.6) with  $N = \infty$ .

In general, this need not necessarily be equivalent to (2.11) in the limit  $N \rightarrow \infty$  but we accept such a specification of  $H^{-1}$  and intend only to accelerate the  $N \rightarrow \infty$  convergence by appropriate modifications of (2.11).

### 3.2. Fixed point approximation

At any finite index  $k = M < \infty$ , merely one of the initialisations (3.2) may be precisely equivalent to the  $N \rightarrow \infty$  limit of (2.11). In its vicinity, there still exists a class of the approximate or 'effective' initialisations which would give the exact result in the  $M \rightarrow \infty$  limit. Finally, in the light of the example given in § 4.2 some of the remaining initialisations may generate, in principle, entirely non-physical inverse matrices.

In practical applications, a construction of the 'effective' initialisation may often be based on the weak  $N$ -dependence of the approximate values (3.2) when evaluated from (2.11) with some finite  $N \gg M$ . Usually, the reliability of such a construction is closely related to the asymptotic smoothness (i.e. slowness of variation or non-oscillatory character) of the elements  $c_M \sim c$ ,  $b_M \sim b$  and  $a_M \sim a$  of  $H$  as functions of their index  $M \gg 1$ .

*Vice versa*, in the smooth cases we may also expect an approximate index-independence of the asymptotic auxiliary sequences (3.2) themselves. Obviously, the constant approximants

$$f_M \sim f \quad u_M \sim u \quad v_{M+1} \sim v \quad \alpha_M \sim \alpha \quad \delta_{M+1} \sim \delta \dots$$

must satisfy the approximate recurrences and may be therefore identified with some of the roots of the system

$$\begin{aligned} \alpha^2 - \alpha u + f &= 0 & \delta^2 - \delta v + f &= 0 \\ u &= (b - v)f & v &= (c - u)f \\ 1/f &= a - (uv/f) - f \end{aligned} \quad (3.3)$$

i.e. with the stationary (fixed) points of the five mappings (2.10)–(2.13).

The algebraic determination of the fixed points may be reduced here to the four definitions

$$\begin{aligned} 2\alpha &= u + (u^2 - 4f)^{1/2} & 2\gamma &= v + (v^2 - 4f)^{1/2} \\ 2\beta &= u - (u^2 - 4f)^{1/2} & 2\delta &= v - (v^2 - 4f)^{1/2} \end{aligned} \quad (3.4)$$

of the asymptotic two-diagonal factors in (2.5), with

$$\begin{aligned} u &= \frac{f}{1+f}B + \frac{f}{1-f}A & v &= \frac{f}{1+f}B - \frac{f}{1-f}A \\ B &= (b+c)/2 & A &= (b-c)/2 \end{aligned} \quad (3.5)$$

and accompanied by the last item of (3.3). When we denote

$$2X = f + 1/f \quad \text{or} \quad f = X - (X^2 - 1)^{1/2}$$

for simplicity, this equation appears to be a cubic one,

$$2X = a - B^2/(2X + 2) + A^2/(2X - 2) \quad (3.6)$$

and defines the last unknown fixed-point parameter in a purely algebraic manner.

The solution of (3.6) is not unique. Examples of analysis of the corresponding 'boundary-type' conditions may be found in I or in § 4.2.

### 3.3. A systematic acceleration of convergence

For the matrices  $H$  which violate the smoothness requirement, we have to reintroduce the indices into the definitions (3.4)–(3.6) ( $a \rightarrow a_M$ ,  $B$ ,  $A \rightarrow (b_M \pm c_{M+1})/2$  and  $X \rightarrow X(M+1)$  in (3.6),  $f \rightarrow f(M+1)$  in (3.5) and  $u \rightarrow u(M)$ ,  $v \rightarrow v(M+1)$ ,  $\alpha, \beta \rightarrow \alpha(M)$ ,  $\beta(M)$  and  $\gamma, \delta \rightarrow \gamma(M+1)$ ,  $\delta(M+1)$  in (3.4)) and consider the deviations

$$U_k = u_k - u(k) \quad V_{k+1} = v_{k+1} - v(k+1) \quad F_{k+1} = f_{k+1} - f(k+1) \quad (3.7)$$

etc, as the new auxiliary sequences. For them, the new recurrences may be derived by simple insertions.

On the basis of the weakened 'smoothness' assumption, we may again introduce the fixed point approximation

$$U_k \sim U(k) \quad V_{k+1} \sim V(k+1) \quad F_{k+1} \sim F(k+1)$$

etc. Its merit is an expected smallness of the roots—we may specify them uniquely from the linearised equations giving, for example,  $F(k+1)$  as a simple rational function of the quantities  $f, f^\Delta, S^2, T^2$  and  $S^\Delta$  where

$$\begin{aligned} f^\Delta &= f(k+1) - f(k+2) \\ u(k) &= S + T \quad v(k+1) = S - T \\ u^\Delta &= u(k) - u(k+1) = S^\Delta + T^\Delta \\ v^\Delta &= v(k+1) - v(k+2) = S^\Delta - T^\Delta. \end{aligned}$$

In principle, the higher-order fixed-point approximants may be generated algebraically in a systematic way. Their shortcoming is their increasingly complicated form—for example, we get (with  $\varphi = f(k+1) + F(k+1)$ )

$$U(k) = \frac{F(k+1)}{f(k+1)} \left( \frac{S}{1+\varphi} + \frac{T}{1-\varphi} \right) + \varphi \left( \frac{S^\Delta}{1+\varphi} - \frac{T^\Delta}{1-\varphi} \right)$$

as the second-order analogue of (3.5), the sixth-order polynomial counterpart to (3.6), etc.

## 4. Applications

### 4.1. Anharmonic oscillator

In the harmonic oscillator basis  $|n\rangle$ ,  $n = 0, 1, \dots$ , some properties of bound states  $\psi$  of the anharmonic Hamiltonian

$$\mathcal{H} = p^2 + x^2 + \lambda x^4$$

may be inferred directly from the Schrödinger equation

$$\mathcal{H}\psi = E\psi$$

or rather from its projection

$$\hat{Q}Q\psi = 0 \quad Q = E - \hat{Q}\mathcal{H}\hat{Q} \quad \hat{Q} = \hat{Q}^2$$

i.e. from the linear relations

$$Q\chi = \varphi \quad \chi = \hat{Q}\psi \quad \varphi = \hat{Q}\mathcal{H}\hat{P} \cdot \hat{P}\psi \quad (4.1)$$

between the 'model-space' projections  $\hat{P}\psi$  ( $\hat{P} = 1 - \hat{Q} = \sum_{m=0}^n |m\rangle\langle m|$ ) and the rest of  $\psi$ . The main reason is that the matrix  $Q$  is five-diagonal; an explicit form of its matrix elements may be found, e.g., in Graffi and Grecchi (1975).

The explicit inversion of  $Q$  in (4.1) may be based on its factorisation (2.5) in the asymptotic region where we may put

$$G_i = D_i = n^2 \times \text{constant}(i) \times O(n) \quad n \gg 1$$

and

$$a_i = a = 6 \quad b_i = c_{i+1} = b = B = 4 \quad A = 0$$

in (2.3). Then, the fixed-point prescription (3.6) degenerates to the quadratic equation with the unique root  $X = f = 1$ . From the remaining relations, we get also  $u = v = 2$ ,  $\alpha = \beta = \gamma = \delta = 1$  and

$$H^{-1} = \begin{pmatrix} 1 & -2 & 3 & -4 & \dots \\ -2 & 5 & -8 & 11 & \dots \\ 3 & -8 & 14 & -20 & \dots \\ \dots & & & & \end{pmatrix} \quad (4.2)$$

(cf (2.6)). Hence, in the leading order approximation, the algebraic inversion of  $Q$  is simple and unique. It also demonstrates that the convergence of  $\|\psi\|$ , if any, is rather slow and may be characterised only in terms of the higher-order corrections. This will not be done here.

#### 4.2. Decadic—decadic propagator

A symmetrically anharmonic phenomenological Hamiltonian

$$\mathcal{H} = \alpha_5 p^{10} + \alpha_4 p^8 + \dots + \alpha_1 p^2 + \beta_1 x^2 + \dots + \beta_4 x^8 + \beta_5 x^{10} \quad \alpha_5 > 0 \quad \beta_5 > 0 \quad (4.3)$$

was introduced by Znojil (1981). In the asymptotic ( $\hat{Q}$ -projected, high-lying harmonic-oscillator) region, it was shown to acquire approximately the five-diagonal form, with

$$G_i = D_i = n^5 \times \text{constant}(i) + O(n^4) \\ a_i = a = 25.2 \quad b_i = c_{i+1} = B = 12 \quad A = 0$$

in (2.3). The numerical inversion of  $H$  gave  $H_{11}^{-1} = 0.055\,728\dots$

In the present ECF context, we shall reproduce this numerical value by purely non-numerical means, clarify its algebraic background and illustrate the character of ambiguities met during its derivation. To achieve this, it is sufficient to apply the results of § 3—the two roots of (3.6) and four values of  $f$  are listed in table 1.



**Table 1.** Numerical values of parameters for the asymptotic factorisation of the decadic-decadic propagators ( $R_{\pm} = 1 \pm 2/\sqrt{5}$ , second row is not spurious).

| $X$            | $f(=H_{11}^{-1})$                | $u$    | $\max(\gamma, \delta)$ | $\min(\gamma, \delta)$ |
|----------------|----------------------------------|--------|------------------------|------------------------|
| 9              | $5R_+^2 (=17.944\ 271\ 90\dots)$ | $6R_+$ | $5R_+ (=19.47\dots)$   | $R_+$                  |
| 9              | $5R_-^2 (=0.055\ 728\ 090\dots)$ | $6R_-$ | $5R_- (=0.527\dots)$   | $R_-$                  |
| $\frac{13}{5}$ | 5                                | 10     | $5R_+$                 | $5R_-$                 |
| $\frac{13}{5}$ | $\frac{1}{5}$                    | 2      | $R_- (=1.894\dots)$    | $R_-$                  |

The second row in table 1 reproduces the numerical solution and coincides, therefore, with the stable fixed point of the corresponding mappings initialised by (2.11) at  $N \rightarrow \infty$ . The remaining items define the spurious solutions and the inverse  $H^{-1}$  which cannot be obtained by the ordinary truncation method.

Without any recourse to numerical computations, the rigorous elimination of the spurious inverses is easy in the present example. Indeed, we get the wavefunction  $\hat{q}\psi$  from (4.1) and (4.3) as a superposition of the first  $n+1$  columns of the matrix

$$\begin{pmatrix} 1 & & \\ -\delta & 1 & \\ \delta^2 & -\delta & 1 \\ \dots & & \end{pmatrix} \times \begin{pmatrix} 1 & & \\ -\gamma & 1 & \\ \gamma^2 & -\gamma & 1 \\ \dots & & \end{pmatrix} = \begin{pmatrix} 1 & & \\ -(\gamma+\delta) & 1 & \\ \gamma^2 + \gamma\delta + \delta^2 & -(\gamma+\delta) & 1 \\ \dots & & \end{pmatrix}. \quad (4.4)$$

Since

$$\delta^m + \delta^{m-1}\gamma + \dots + \gamma^m = (\delta^{m+1} - \gamma^{m+1})/(\delta - \gamma) \quad \delta \neq \gamma \quad (4.5)$$

we arrive at the rigorous asymptotic estimate

$$| \langle m+1 | \chi \rangle / \langle m | \chi \rangle | = \max(|\delta|, |\gamma|) \quad m \gg n \geq 0. \quad (4.6)$$

Now, the geometric convergence criterion (Korn and Korn 1968) implies that none of the spurious fixed points  $f$  can lead to the normalisable eigenstate  $\psi$  or  $\chi$  of the decadic-decadic Hamiltonian  $\mathcal{H}$  since  $\max(\gamma, \delta) > 1$ .

## 5. Generalisations and summary

Any band matrix  $Q$  with  $t$  upper and  $s$  lower diagonals may be factorised in analogy with the  $t=s=2$  example of § 2. Similarly, even to invert any blockwise Hessenberg matrix with the variable partitions, the same decomposition into the two-diagonal matrix factors with some zero elements  $\alpha_i, \dots$  may be used. The resulting inverses will be generalisations of equation (2.6).

A peculiarity of the three- and five-diagonal examples lies in their non-numerical character (cf our ECF theorem, the formulae in the appendix or the simplicity of the fixed points). In the applications, they may therefore be used as the solvable models of interaction (see our first example) etc.

The pentadiagonal extension of the tridiagonal matrices may also prove to be useful as the mathematical approximant whenever the ordinary methods diverge (cf, e.g., the perturbative expansion of our first example as discussed, e.g., by Killingbeck

(1977)), or oscillate ('averages' of Richardson and Blankenbecler (1979) resemble our fixed points). In numerical practice, they could also sometimes replace the standard (finite-submatrix) truncation approximants in a way similar to our second example.

### Appendix. Compactified form of inverse of the five-diagonal matrix $H$

The explicit form (2.6) of  $H^{-1}$  contains the four sequences  $\alpha_k$ ,  $\beta_k$ ,  $\gamma_{k+1}$  and  $\delta_{k+1}$ . The main idea of their replacement by mere 'sums'  $u_k$  and  $v_{k+1}$  (compare equations (2.14) and (2.17)) lies in the direct use of the 'upper times lower' decomposition (2.5) written in the form

$$H = \begin{pmatrix} 1 & u_1 & f_3 & & \\ & 1 & u_2 & f_4 & \\ & & \dots & & \\ & & & 1 & u_{N-1} \\ & & & & 1 \end{pmatrix} \times \begin{pmatrix} 1/f_1 & & & & \\ & \ddots & & & \\ & & \ddots & & \\ & & & \ddots & \\ & & & & 1/f_N \end{pmatrix} \times \begin{pmatrix} 1 & & & & \\ v_2 & 1 & & & \\ f_3 & v_3 & 1 & & \\ & \dots & & \ddots & \\ \dots & f_N & v_N & 1 \end{pmatrix}. \quad (\text{A1})$$

We shall now construct  $H^{-1}$  by inverting the triangular factors in (A1),

$$H^{-1} = \begin{pmatrix} 1 & & & & \\ z_2^{(1)} & 1 & & & \\ z_3^{(1)} & z_2^{(2)} & 1 & & \\ \dots & & & \ddots & \\ z_N^{(1)} & \dots & & & 1 \end{pmatrix} \times \begin{pmatrix} f_1 & & & & \\ f_2 & & & & \\ & \ddots & & & \\ & & \ddots & & \\ & & & \ddots & f_N \end{pmatrix} \times \begin{pmatrix} 1 & y_1^{(1)} & y_2^{(1)} & \dots & y_{N-1}^{(1)} \\ & 1 & y_1^{(2)} & \dots & y_{N-2}^{(2)} \\ & & \ddots & \ddots & \\ & & & \ddots & \\ & & & & 1 \end{pmatrix}. \quad (\text{A2})$$

Such an approach cannot be generalised too easily but it gives a more compact result—the multiplication in (A2) is still feasible 'by hand' for the five-diagonal  $H$ . We intend to show here how the necessary algebraic manipulations may employ the recurrences valid for the sequences  $u_k$ ,  $v_{k+1}$  and  $f_k$ .

Without loss of generality, we shall evaluate only the sequence  $z_k^{(1)}$ ,  $k = 2, 3, \dots, N$ . From its definition we get, omitting the upper indices,

$$v_2 + z_2 = 0 \quad f_3 + v_3 z_2 + z_3 = 0 \dots$$

i.e. with  $z_1 = 1$  and  $z_0 = 0$ ,

$$z_k = -v_k z_{k-1} - f_k z_{k-2} \quad k = 2, 3, \dots, N. \quad (\text{A3})$$

The explicit form of the first few expressions  $z_k$  inspires us to use the ansatz

$$z_k = f_2 f_3 \dots f_k [A_k + f_{k+1} (B_k + D_k v_{k+2})] \quad (\text{A4})$$

$$A_1 = -D_2 = 1 \quad B_1 = D_1 = 0 \quad A_2 = -c_2 \quad B_2 = b_2 \dots$$

In this setting, the mathematical induction is easy (cf also the proof of the theorem in § 2) and gives the recurrences

$$D_k = -A_{k-1} \quad B_k = A_{k-2} + A_{k-1} b_k \quad (\text{A5})$$

$$A_k = -(B_{k-2} + a_{k-1} A_{k-2} + c_k A_{k-1}) \quad k = 3, 4, \dots$$

Obviously, the first two of them are mere definitions while the third one,

$$A_k = -(c_k A_{k-1} + a_{k-1} A_{k-2} + b_{k-2} A_{k-3} + A_{k-4}), \quad (\text{A6})$$

has an explicit solution

$$A_k = (-1)^{k+1} \det S(k) \quad k = 2, 3, \dots, N \quad (\text{A7})$$

$$S(k) = \begin{pmatrix} c_1 & a_2 & b_2 & 1 & 0 & \dots \\ 1 & c_3 & a_3 & b_3 & 1 & 0 & \dots \\ & \dots & & & & & \\ 0 & \dots & & 0 & 1 & c_{k-1} & a_{k-1} \\ 0 & \dots & & & 0 & 1 & c_k \end{pmatrix}.$$

After a simple rearrangement of (A4), we get the final formula

$$z_k^{(1)} = (-1)^{k+1} f_2 f_3 \dots f_k [\det S(k) - u_k \det S(k-1) + f_{k+1} \det S(k-2)] \quad (\text{A8})$$

$$k = 2, 3, \dots, N \quad \det S(0) = 0 \quad \det S(1) = 1.$$

Its main merits are:

(i) When inserted into (A2), it provides a compact definition of  $H_{ij}^{-1}$  for the small indices  $i \neq j \ll N$ .

(ii) Due to its linear  $u$ -dependence (which simplifies also the transpositions  $H(u, v) = H^T(v, u)$ ), its ECF representation is very similar to the ECF denominator itself.

(iii) The presence of factors  $f_i f_{i+1} \dots f_{i+k}$  in  $H_{ii+k}^{-1}$  resembles and generalises the continued-fraction factorisation encountered in the tridiagonal case (see I) and may also be generalised to the more-diagonal matrices.

The diagonal matrix elements  $H_{ii}^{-1}$  are exceptional. Their ECF form is extremely simple

$$H_{ii}^{-1} = f_1 f_2 \dots f_i [\det H(i-1) - f_{i+1} \det H(i-2)]$$

$$i = 1, 2, \dots \quad \det H(-1) = 0 \quad \det H(0) = 1 \quad (\text{A9})$$

$$H(i) = \begin{pmatrix} a_1 & b_1 & 1 & \dots \\ \dots & & & \\ 0 & \dots & 1 & c_i & a_i \end{pmatrix}$$

and follows directly from (A1) and from the Kramer rule.

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